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Lab 7: Programs on Strings in CUDA

Q1).
Code:

```
#include <stdio.h>
#include <string.h>

__global__ void countWord(char *sentence, char *word, int *count, int slen, int wlen) {
    int idx = blockIdx.x * blockDim.x + threadIdx.x;

    if (idx <= slen - wlen) {
        int match = 1;
        for (int i = 0; i < wlen; i++) {
            if (sentence[idx + i] != word[i]) {
                match = 0;
                break;
            }
        }
        if (match)
            atomicAdd(count, 1);
    }
}

int main() {
    char sentence[200], word[50];
    int *d_count, count = 0;
    char *d_sentence, *d_word;

    printf("Enter sentence: ");
    fgets(sentence, 200, stdin);
    printf("Enter word to search: ");
    scanf("%s", word);

    int slen = strlen(sentence);
    int wlen = strlen(word);
    cudaMalloc(&d_sentence, slen);
    cudaMalloc(&d_word, wlen);
    cudaMalloc(&d_count, sizeof(int));

    cudaMemcpy(d_sentence, sentence, slen, cudaMemcpyHostToDevice);
    cudaMemcpy(d_word, word, wlen, cudaMemcpyHostToDevice);
    cudaMemcpy(d_count, &count, sizeof(int), cudaMemcpyHostToDevice);

    countWord<<<(slen+255)/256, 256>>>(d_sentence, d_word, d_count, slen, wlen);
    cudaMemcpy(&count, d_count, sizeof(int), cudaMemcpyDeviceToHost);
    printf("Word repeated %d times\n", count);
    cudaFree(d_sentence);
    cudaFree(d_word);
    cudaFree(d_count);
    return 0;
}
```

Output :

```
STUDENT@MIT-ICT-LAB5-15:~/Desktop/230905408/lab7$ ./q1
Enter sentence: 1 + 1 + 2 + 3 + 2+ 8 + 9
Enter word to search: +
Word repeated 6 times
STUDENT@MIT-ICT-LAB5-15:~/Desktop/230905408/lab7$
```

Q2)

Code:

```
// string_RS
#include <stdio.h>
#include <string.h>

__global__ void generateRS(char *S, char *RS, int N) {
    int idx = threadIdx.x;

    if (idx < N) {
        int start = idx * N - (idx * (idx - 1)) / 2;
        for (int i = 0; i < N - idx; i++) {
            RS[start + i] = S[i];
        }
    }
}

int main() {
    char S[100];
    char *d_S, *d_RS;

    printf("Enter string S: ");
    scanf("%s", S);

    int N = strlen(S);
    int RS_len = N * (N + 1) / 2;

    char *RS = (char*)malloc(RS_len + 1);

    cudaMalloc(&d_S, N);
    cudaMalloc(&d_RS, RS_len);

    cudaMemcpy(d_S, S, N, cudaMemcpyHostToDevice);

    generateRS<<<1, N>>>(d_S, d_RS, N);

    cudaMemcpy(RS, d_RS, RS_len, cudaMemcpyDeviceToHost);
    RS[RS_len] = '\0';

    printf("Output RS: %s\n", RS);

    cudaFree(d_S);
    cudaFree(d_RS);
    free(RS);
    return 0;
}
```

Output:

```
● STUDENT@MIT-ICT-LAB5-15:~/Desktop/230905408/lab7$ ./q2
  Enter string S: helloworld
  Output RS: helloworldhelloworlhelloworhellowohellowhellohellhelheh
○ STUDENT@MIT-ICT-LAB5-15:~/Desktop/230905408/lab7$ 
```